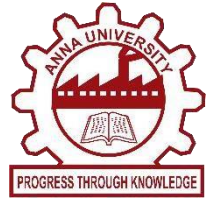




PATIENT MONITORING SYSTEM USING RASPBERRY PI AND GSM MODULE



A MINI PROJECT REPORT

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ABSTRACT

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Real time monitoring of patients is critically important in modern healthcare because intervention must be timely and outcomes improved. In this project we propose a low cost reliable patient monitoring system which will be monitored with the help of Raspberry Pi micro controller and GSM module to monitor two vital parameters. heart rate and body temperature. Biomedical sensors collect data in real time which can then be processed by the Raspberry Pi. A pulse sensor measures heart rate while a digital temperature sensor measures body temperature. When the reading is abnormal, the system sends an SMS alert through the GSM module to the registered caregiver or medical personnel, and it is possible to respond promptly without direct supervision.

Considering affordability and portability, this system is ideal for low resource setting such as rural clinics, home-based care and emergency. Unlike the hospital-grade monitors of the past, it is compact, light weight, does not need internet connectivity, which makes it suitable for use outside the hospital. Its easy user-friendly interface allows non-technical users such as family caregiver or community health worker to use the computer. Modular design permits further expansion with another sensor or wireless capabilities, including Wi-Fi or Bluetooth. This project is a demonstration of the capabilities of the embedded system and mobile communication, to provide accessible real-time patient monitoring and enhance healthcare in underserved areas.

KEYWORDS: Patient Monitoring, Raspberry Pi, GSM Module, Heart Rate Sensor, Temperature Sensor, Real-Time Alerts, Embedded Systems, Remote Healthcare, Low-Cost Medical Device, Emergency Response.

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LIST OF ABBREVIATIONS

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ACRONYMS

GSM

SMS

LCD

CNN

ECG

GPIO

GPRS

HDMI

ADC

RAM

BPM

ABBREVIATIONS

**Global System for Mobile
Communications**

Short Message Service

Liquid Crystal Display

Convolutional Neural Network

Electrocardiogram

General Purpose Input/Output

General Packet Radio Service

High-Definition Media Interface

Analog to Digital Converter

Random Access Memory

Beats Per Minute

CHAPTER 1
INTRODUCTION

CHAPTER 1

INTRODUCTON

A patient monitoring system based on a Raspberry Pi and a GSM module is smart and inexpensive way to monitor vital health signs in real time. As the brain of the system, the Raspberry Pi picks up information from sensors that monitor such things as heart rate, body temperature, and oxygen levels. This data can be viewed immediately or saved for future reference, which can be particularly useful in areas where it isn't easy to access healthcare regularly – those remote or homebound patients.

The most useful aspect is the GSM module which allows the sending of instant text alerts. If there is any reading outside the normal range, the system will immediately send an SMS to a caregiver or a doctor. This quick notification can be a lifesaver especially for the elderly, long term illnesses or any individual recovering from surgery who needs special care at home.

Apart from live monitoring, the system can also log data over time and connect to the cloud or mobile app to access it from afar. It's also flexible you can add more sensors if need be. Also, it's energy efficient and can even function off of batteries or solar power, making it a great choice for low resource/rural settings. Overall, it's a significant move towards making healthcare accessible, and responsive.

1.1 BACKGROUND

As the demand for continuous and remote monitoring of health particularly for the seniors, those that have chronic conditions as well as those in difficult to access areas has risen, healthcare technology has had to rise too. Hospital monitors traditionally are useful but costly and are stationary, so it's difficult to use them outside clinics. This is why there is a growing interest in the portable health

monitoring systems that are not expensive, and the one that works in real time – from a person’s home without constant doctor’s attention.

That’s where the Raspberry Pi comes into the battle. It is an inexpensive computer that is perfect for building such system. It can connect to sensors that track such vital signs as heart rate, body temperature and oxygen.

Besides, the system will also be able to store the data, send it to the cloud, or even integrate with a mobile app for access from anywhere. It is also flexible, additional sensors are able to be added for ECG, blood pressure or glucose monitoring etc. Since it does not use a lot of power, it can be powered from batteries or solar energy thus a good fit for such areas that are resource challenged.

1.2 NEED FOR THE PROJECT

As people are more and more living with chronic illnesses while the number of older people increases, the need for efficient and easily accessible patient monitoring systems increases. Most of the traditional healthcare setups find it difficult to offer constant care – this especially in rural or remote areas, where it can take hours before reaching a clinic or hospital. Even normal check up can be hard for people in these regions, so that warning signs will be missed and treatment delayed. This is why there is such a need for a simple, inexpensive solution that will allow them to monitor life signs and catch problems early without the need for the patients to constantly visit a hospital.

A patient monitoring system constructed with a Raspberry Pi and a GSM module solves this problem. It’s inexpensive, portable, and easy to scale. Pi is sort of a mini computer that collects data from the sensors that measure the important health indicators such as heart rate, body temperature and oxygen. If anything is at all out of the ordinary, the system immediately alerts a caregiver or healthcare provider by

text through the GSM module – so that help can be provided to the patient as soon as possible. Such a real time response can be life saving in emergencies.

1.3 PROBLEM IDENTIFICATION

In most parts of the world – in rural or remote areas, in particular – accessing regular, decent healthcare is still quite a problem.

When people live with chronic conditions or must be regularly checked, they often don't have the time, the money, or the nearby facilities to obtain the care necessary. They are expensive, and they require constant medical supervision and are simply impractical to be used in the home environment or in a low-resource setting.

This problem could be solved in a smart and cheap way using the Raspberry Pi with a GSM module. But there are still barriers to overcome – such as ensuring the accuracy of the sensors, the reliability of the data transmission, the consumption of power by the system not being excessive and the ease of use of the interface by anyone. We should also ensure confidentiality of patient data, especially if it's stored or shared online. Because of unreliable internet connection in remote areas, the system must also function properly offline. And for it to actually make a difference, it should be easy to install, easy to maintain and built from inexpensive readily available parts.

1.4 OBJECTIVE

The primary goal of the patient monitoring system that incorporates the use of Raspberry pi and GSM module is to develop an affordable, effective and real time health monitoring system to support constant monitoring of vital signs from outside the standard hospital environment. This system is tailored to ensure improved access to healthcare in the rural or remote areas where early detection of health related

problems can be done, and hence medical intervention can be implemented immediately.

Key Objectives:

- **Monitor Vital Health Parameters:** Use biomedical sensors that will be attached to the Raspberry Pi which will monitor the most important health parameters, such as heart rate, body temperature, and oxygen saturation.
- **Real-Time Data Processing:** Run real time process and analysis for sensor data to detect any anomaly or health crisis as it occurs.
- **Immediate alerts:** When abnormal readings show up SMS alerts the healthcare professionals directly through a GSM module which requires further action.
- **User-Friendly and Portable:** Make the system easy to use, portable, convenient for home and remote usage.
- **Data Logging and Remote Access:** Request long term data tracking and allow remote monitoring by mobile apps for continuous health management.
- **Energy Efficiency:** Make the system energy efficient, battery or solar powered, so that it works well where there is poor access to electricity.
- **Scalable and Modular:** Develop a flexible system that could be extended with new sensors for different medical uses – as it was thought required.
- **Improved Healthcare Delivery:** The system is designed to improve healthcare through available efficient and timely monitoring leading to quicker and well informed decision.

1.5 ORGANIZATION OF PROJECT

The project's fundamental structure is as follows:

CHAPTER 1: This chapter deals with the Introduction and the overview to have a basic idea about the project.

CHAPTER 2: This chapter deals with Literature Survey for better understanding and relevance for the enhancement of the proposed work.

CHAPTER 3: This chapter describes the existing work and the technology used in the existing project.

CHAPTER 4: This chapter describes the component work and the technology used to improve the project.

CHAPTER 5: This chapter deals with the software and hardware descriptions which are used in this technology.

CHAPTER 6: This chapter deals with the result and discussion of the project

CHAPTER 7: This chapter explains the conclusion which is obtained and scope for future work on the project.

1.6 SUMMARY

This patient monitoring system created using Raspberry Pi and GSM module is a smart and pocket friendly way of monitoring somebody's health. It tracks vital body signs like heart rate, body temperature, and oxygen levels in real time using sensors. In case there are strange things that emerge, the system immediately sends a SMS alert to family members or healthcare givers, so that they can act in time.

It is especially useful for those who live in outback or need regular routines at home. The system is user-friendly, energy-saving and portable – it is therefore fairly

appropriate even where power or medical facilities are scarce. It also maintains health data to be used in the future and can attach to cloud services or mobile applications hence easy to access from any location. In addition, it is possible to add additional sensors if needed and develop it into a flexible and reliable tool to help facilitate better, quicker, and more accessible healthcare.

CHAPTER 2
LITERATURE REVIEW

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

As healthcare keeps on changing, technology is getting more and more important in the ways that we take care of our patients, particularly in monitoring their health outside the hospitals. As more people require frequent health check-ups, especially for the chronically ill or remote dwelling, there is a need for smart affordable solutions that enable one to do this at home. The Raspberry Pi, a tiny but inexpensive computer supplemented by a GSM module for mobile communication, has become a favorite option for such systems.

In this review, we examine the ways in which researchers and developers are applying Raspberry Pi and GSM technology in patient monitoring projects. It discusses the various applications in which sensors are being used, how health data is shared, and the roadblocks that have been overcome. By examining the current work that already exists, we will be able to see what has been achieved, and we will know how we can improve the future systems to make them more reliable, user-friendly and effective for real world healthcare requirements.

2.2 HARDWARE-BASED CONVOLUTIONAL MODEL FOR THE EARLY IDENTIFICATION OF HEALTH DISEASES:

Convolution hardware based early disease detection model uses deep learning, which is Convolutional Neural Network (CNNs) to analyze and detect health conditions in the real-time. Against a backdrop of growing demand for convenient and readily available healthcare, especially in outlying or under-served areas, this method represents an attractive alternative to traditional cloud-based options.

This arrangement operates locally on data and therefore does not need an internet connection or powerful servers. It not only shortens the response times; it also facilitates protection of patients' privacy by ensuring that sensitive information does not travel over the networks. For example, a wearable device's ECG signal can be read by the onboard CNN model immediately and it can warn if any irregularities are identified like heart conditions. Similarly chest images can be checked early for the presence of pneumonia or COVID-19, thus it becomes possible to intervene in a timely fashion. The ability to move such systems makes them perfect for use in ambulances or in home care or rural clinics. Additionally, their low power consumption and cost effectiveness would make them scalable for large healthcare deployments. With the AI incorporated with the hardware this solution helps to close the gap between the smart diagnostics and accessibility enhancing the healthcare more proactive, personalized and responsive.

2.3 EARLY DETECTION OF HEALTH DISEASE USING CONVOLUTIONAL HARDWARE COMPONENT.

Timeous detection of health problems is critical to effective treatment, better results and, cost of healthcare disease. Application of CNN models on hardware for monitoring patients makes real time health analysis more practical and effective. This project proposes the use of raspberry pi as a cheap processing unit and use of a gsm module for remote connection and lightweight cnn models that can be able to detect early signs from the sensor data of health problems. These models are taught to observe patterns in such basic signs as ECG signals, heart rate, body temperature, and oxygen levels in order to identify issues such as arrhythmia, infections, or breathing, The CNN model is embedded directly into the Raspberry Pi and is optimized to operate quickly and locally without a cloud-based processing.

This system provides a cheap, portable and smart healthcare solution in a combination of convolutional AI and embedded hardware. It ensures proactive monitoring, reduced hospital visits and ensures quick action when needed. Ultimately, this strategy helps create smarter, more accessible and more reactive healthcare.

2.4 MOBILE-BASED DEEP LEARNING MODELS FOR HEALTH DISEASE DETECTION:

Mobile based deep learning models are creating a huge impact in the digital healthcare environment by improving our ability to detect diseases and monitor patients in our ever changing modern world. This strategy combines ease of smartphones with the strength of deep learning, thus making real-time processing of health data possible. In this set up the Raspberry Pi serves as the primary processor with sensors attached to monitor basic signs including heart rate, body temperature and oxygenation. The data that it collects may be processed on the device or sent to a mobile app that runs a trained deep learning model that will identify pre symptoms of potential health problems.

The specific thing that makes this set up especially useful is that there is no need for a constant connection to the internet. This means availability of help will be available shortly – even in remote or under-served regions without connectivity.

The system offers a mobile, easy to use and responsive way of monitoring health using the combination of edge computing to mobile AI. It gives more control to users over their wellbeing, it is able to detect problems early and it closes the gap between home care and professional medical care.

2.5 SUMMARY

According to the literature review, the use of Raspberry Pi and GSM modules in the patient monitoring systems is becoming popular, and for a good reason. These setups are cheap, portable and they can give real time health monitoring which makes them ideal for areas where health care is limited (remote or rural areas). The Raspberry Pi is the main processor and it issues alerts when there is a problem using GSM module. They can monitor important vital signs, such as the heart rate, temperature and oxygen levels and hence be in a position to detect possible health problems at an early stage.

Researchers have observed that these systems are not only energy efficient but also user-friendly and their modular nature is such that one can easily attach more sensors to them and connect them to mobile apps and cloud platforms. However, there is still a job to be done – accuracy does need to be improved and data is being managed, but these systems have a lot of potential in terms of early disease detection and it ensures continuous care of patients irrespective of where they operate.

CHAPTER 3
METHODOLOGY

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

The project idea is now quite clear after a thorough examination of books, magazines, journals, websites, and other resources. The goal is to create a patient monitoring system based on Raspberry Pi and GSM module for collecting, processing, and transmitting real-time health data. The system controller, the Raspberry Pi, interfaces with biomedical sensors including pulse oximeters, temperature sensors, and heart rate sensors. It keeps collecting and analyzing the data to find any unusual patterns in a patient's vital signs.

Should there be an anomaly, the GSM module notifies a doctor or family member via SMS message so that prompt action may be taken—be it in internet-covered or non-internet-covered areas. Designed low-power and portable, it is therefore perfect for use in the field or at home. Every effort is made to make it affordable, exact, and dependable. The solution usually enables fast intervention for possible medical issues and round-the-clock health monitoring, therefore enhancing care where and when it is most needed.

3.2 DATA USED

The patient monitoring system created from Raspberry Pi and GSM technology primarily gathers real-time physiological signals from biomedical sensors.

The key types of data used include:

Heart Rate (BPM) is determined through pulse sensors or heart rate monitors which help identify irregular patterns such as tachycardia or bradycardia.

Body Temperature The system acquires body temperature data through digital

temperature sensors such as DS18B20 or LM35 to monitor potential cases of fever or hypothermia.

Time Stamps function by logging each data point with a time stamp which allows for the monitoring of trends and patterns throughout time.

Alert Status The system uses Boolean values or status indicators to determine if readings are normal or abnormal which activates the GSM alert system.

The Raspberry Pi processes all data locally to assess whether it needs to activate the GSM module for sending health alerts to caregivers or healthcare providers.

3.3 BLOCK DIAGRAM OF PATIENT MONITORING SYSTEM:

This block diagram represents a real-time health monitoring system that uses a Raspberry Pi as its central controller. The system is designed to keep track of vital health parameters and instantly alert caregivers or medical personnel when necessary

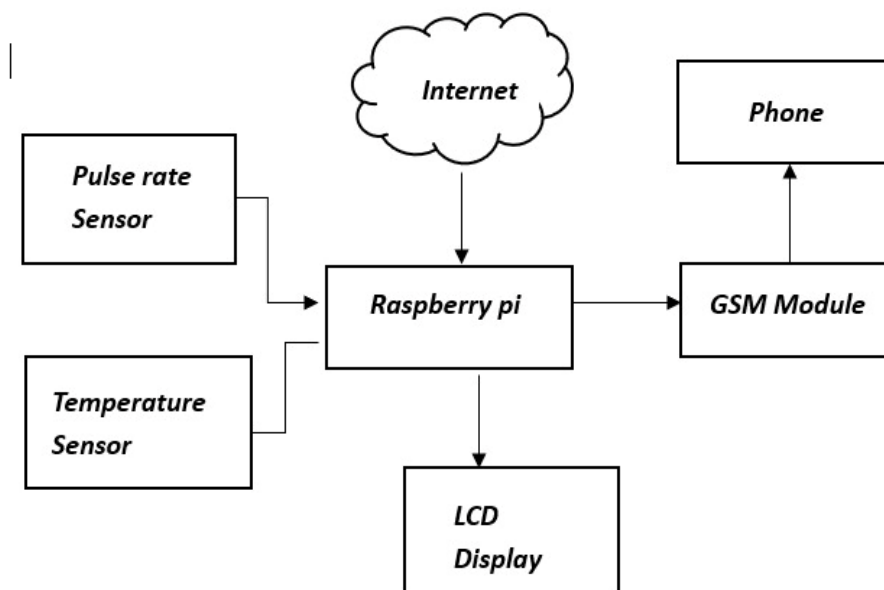


Fig 3.1 Block diagram of Patient Monitoring system

Pulse Rate Sensor:

This sensor continuously monitors the heart rate of the patient and sends the data to the Raspberry Pi.

Temperature Sensor:

It measures the body temperature and also sends the readings to the Raspberry Pi.

Raspberry Pi:

Acting as the brain of the system, the Raspberry Pi collects data from the sensors, processes it, and takes appropriate action based on predefined conditions.

LCD Display:

The real-time values of pulse rate and body temperature are displayed on an LCD screen for easy monitoring by the user or caregiver.

Internet Connectivity:

The Raspberry Pi connects to the internet, which can be used to upload data to cloud platforms for remote monitoring and long-term health tracking.

GSM Module:

If any abnormal readings are detected, the Raspberry Pi sends an alert through the GSM module, which can call or text a predefined emergency contact (like a family member or doctor).

Phone:

The alert is received on a mobile phone, allowing timely medical intervention if needed.



DHT11 SENSOR: Measures temperature and humidity.

GSM MODULE: (e.g. SIM900) Sends SMS alerts.

ADC(MCP3008): Converts analog signals from sensors into digital form for the Raspberry Pi. It enables interfacing analog medical sensors with the Pi, which lacks built-in ADC support.

3.5 IMPLEMENTATION:

Home-Based Patient Care:

This system is perfect for keeping an eye on elderly or chronically ill family members. It sends instant alerts to caregivers or doctors if something goes wrong, making sure help is never too far away when it's needed most.

Rural and Remote Healthcare:

In places where healthcare facilities are far away or the internet is spotty, this system can still work perfectly. It uses GSM to send vital health data directly to doctors, no matter how far away they are, helping bridge the healthcare gap in rural areas.

Hospitals and Clinics:

Smaller clinics or even hospital wards can use this system to monitor patients without needing expensive or complex equipment. Nurses and medical staff get instant alerts by text when a patient's condition suddenly worsens, making it easier to act quickly.

ICU and Post-Operative Monitoring:

This system will guarantee continuous monitoring of vital signs in high risk settings including ICUs and recovery areas. In the event of a problem, a rapid alert is issued which allows rapid response from the medical team, a critical element in emergencies.

COVID-19 Isolation Wards:

Pandemic, medical staff safety was critical. This system allows the healthcare workers to monitor the patients of COVID without having to go face-to-face with

them thus reducing the consumption of necessary personal protective equipment such masks and gloves.

Mobile Health Units:

A patient's health information can be sent to the hospital from an ambulance or mobile health unit before patients even get to the hospital. By this, hospitals are able to get organized before patients turn up and have staff on standby.

Wearable and Portable Health Systems:

Some small changes would make this system mobile and flexible for a wear as a device. Consider a person with heart disease or epilepsy wearing the system because it gives constant health control, and it will send a signal if something is wrong.

Elderly Care Facilities and Nursing Homes:

Respond For facilities dealing with the elderly, this system will enable constant monitoring of residents to take action immediately if there is any health concern. When a major event occurs, caregivers are alerted quickly so that they can respond in good time.

Disaster and Emergency Camps:

Almost always with a disaster, people have limited healthcare options. The healthcare providers can install such a system in shelters, which will enable the healthcare providers to keep track of individuals well-being, and even deliver essential care when resources are minimal.

Military and Adventure Medical Support:

For soldiers in remote areas from base and people who go to remote wilderness, this system enables nearby medical teams to get health information instantly through GSM. So, with this, medical aid can get to them fast, even under the harshest conditions.

3.6 SUMMARY:

This patient monitoring system uses a Raspberry Pi to monitor vital signs, including heart rate and body temperature, very closely. The system always checks and reviews this information looking for indications of abnormality. If any alarming signs develop, the device automatically sends a short message alert to the relevant medical professional using the GSM module that is built into the device, without the need for an active internet connection. This will make the system especially useful for the monitoring of patients in isolated or thinly populated areas with poor internet connections.

A strong and reliable power system is in place with backup power solutions for emergencies cases. This means the system is easy to transport or even be worn and this will enable patients to be monitored at all times. It provides real time health tracking at an affordable cost making it both easy to use and suitable for both patients and caregivers.

CHAPTER 4

HARDWARE REQUIREMENTS

CHAPTER 4

HARDWARE DESCRIPTION

4.1 RASPBERRY PI:

The Raspberry Pi 3 Model A+ is a small, efficient, and budget-friendly single-board computer that's increasingly being used in healthcare, especially for patient monitoring. It features a 64-bit quad-core ARM Cortex-A53 processor running at 1.4 GHz, along with built-in Wi-Fi (802.11ac), Bluetooth 4.2, and 512 MB of RAM. This combo makes it perfect for lightweight, embedded tasks where space and power are limited—making it a solid choice for portable or remote health monitoring setups.

In typical patient monitoring setups, the Raspberry Pi 3 Model A+ acts as the main control hub, connecting with various medical sensors through GPIO pins or USB ports. These sensors usually include ones for heart rate and SpO₂ (like MAX30100), temperature sensors (like DS18B20), and ECG modules (such as AD8232). The Pi gathers, processes, and can wirelessly send patient data to cloud servers or mobile apps.

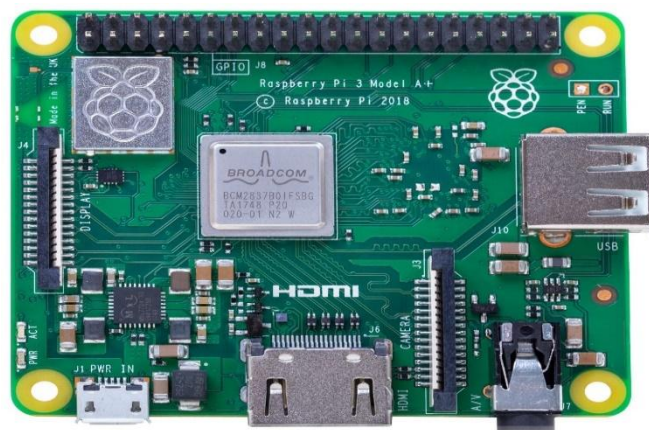


Fig 4.1 Raspberry pi 3 A+

Its small size and low power needs, it's especially useful in rural clinics, elder care homes, or even in patients' homes for recovery. Its built-in wireless capabilities let it easily transmit data without extra modules, keeping the system simple and affordable.

4.2 SPECIFICATION

4.2.1 Processor

At the core of the Raspberry Pi 3 Model A+ is the Broadcom BCM2837B0 SoC, which comes equipped with a 64-bit quad-core ARM Cortex-A53 processor running at 1.4 GHz. While it's not as powerful as the newer Cortex-A72 chip found in the Raspberry Pi 4, this processor still handles most embedded tasks quite well. It's capable of managing sensor data collection, simple image processing, light web browsing, and running headless servers. The four cores ensure smooth multitasking and make real-time operations, like patient monitoring, robotics, or smart home systems, run more efficiently.

4.2.2 RAM

The A+ model has 512 MB of LPDDR2 SDRAM — a bit less than the 1 GB in the 3B+ version. But for many projects that don't need a full graphical interface or heavy multitasking, this is enough. Whether you're working on IoT setups, industrial controls, or medical monitoring systems with lightweight Linux distros like Raspberry Pi OS Lite or custom builds, this memory is reliable. Developers often optimize their apps to use minimal resources, so the 3A+ can handle background tasks effectively without much fuss.

4.2.3 Graphics

The device sports the Broadcom Video Core IV GPU, supporting OpenGL ES 2.0 and 1080p H.264 video decoding. While it's not meant for high-end gaming or intense graphics, it's plenty capable for simple user interfaces, video playback, or small dashboards. Developers can use this GPU to create basic graphical feedback or lightweight Human-Machine Interface (HMI) applications. It outputs video through a standard full-size HDMI port, supporting resolutions up to 1080p.

4.2.4 Connectivity

One of the best features of the Raspberry Pi 3 Model A+ is its connectivity options. It includes:

- Dual-band Wi-Fi (802.11 b/g/n/ac) on 2.4 GHz and 5 GHz, for quick, stable wireless connections, making it ideal for IoT and real-time apps.
- Bluetooth 4.2 and BLE , perfect for linking wireless sensors, peripherals, or mobile devices especially handy in wearable health tech.
- A single USB 2.0 port for peripherals like keyboards, USB drives, or serial connections. If needed, additional ports can be added with powered USB hubs.
- A 40-pin GPIO header, compatible with earlier models, for connecting sensors, actuators, or other custom hardware. It's essential for prototyping or hardware projects.
- It doesn't include Ethernet or USB 3.0 ports, emphasizing its focus on being a small, wirelessly-connected device.

4.2.5 Storage

The main storage method is via a microSD card slot, which houses the OS and files. It supports high-speed cards (Class 10 and above recommended). Swapping or

upgrading your software is straightforward. Unlike the Pi 4, the 3A+ doesn't support native USB boot, but you can still use USB storage devices through the USB 2.0 port for data logs or extra storage if needed.

4.2.6 Power Supply

The board is powered through a standard 5V micro-USB connector, with a recommended 2.5A power supply for stable operation. Thanks to its compact design and fewer components, it's more energy-efficient than larger Pi models. This makes it a good choice for battery-powered setups, remote monitoring stations, or solar-powered projects where saving energy matters.

4.2.7 Other Features

- CSI camera port: Allows you to connect official Raspberry Pi Camera Modules for image capture, useful in surveillance, machine vision, or health monitoring.
- DSI display port: Supports touchscreen displays, enabling interaction with graphical interfaces without external monitors.
- Audio output: Audio can be played through HDMI or via the 3.5mm TRRS jack, which combines analog audio and composite video. Ideal for multimedia projects or simple speaker setups.

4.3 GSM MODULE:

A GSM module is a helpful object for electronic systems as it allows them to work together with mobile networks, the same way that a smartphone can send messages or call. Patient monitoring systems use this module to alert doctors or caregivers by SMS in a timely manner, so support is readily available in the event of emergency readings.



Fig 4.2 GSM Module (SIM 900)

Among the most popular SIM800 and SIM900 modules are compatible with 2G networks and work almost everywhere. You can connect to such devices as the Raspberry Pi using these modules by attaching to them using simple AT commands.

Once a SIM card has been inserted and an antenna attached, the module is ready to send texts, make calls, or transmit brief data packets on the mobile network.

GSM modules, because of their low costs and availability to the general public without the need for internet, are particularly suitable for health tracking for remote areas or areas without internet. The reliable and simple nature of GSM modules make them a feasible choice to achieve functional, real-time healthcare systems.

4.4 SPECIFICATIONS

4.4.1 Network Support

The GSM 900 module operates on quad-band GSM/GPRS frequencies: 850 MHz, 900 MHz, 1800 MHz, and 1900 MHz. This essentially means it can function in almost any part of the world where 2G cellular networks are present. Among these

frequencies, the 900 MHz band is especially effective in areas like Europe and Asia, thanks to its excellent range and ability to penetrate buildings. It's a great choice for remote or mobile applications where there's not always a steady internet connection, but access to the GSM network is good enough.

4.4.2 Data Transmission

When it comes to mobile internet applications, the module is capable of handling:

GPRS multi-slot class 10, which supports moderate data speeds that work well for basic tasks like uploading data to a cloud server or getting configuration updates.

- GPRS mobile station class B, which allows the device to use both circuit-switched and packet-switched services at the same time
- While it may not be as fast as 3G or 4G networks, GPRS is dependable and suitable for sending sensor data or regular status updates without an issue.

4.4.3 Communication Interface

The module connects with external devices through a UART interface, which is a common serial protocol. This makes it a great fit for microcontrollers and platforms like the Raspberry Pi. Developers can easily control the module using AT commands (Attention commands), which are a straightforward text-based command set for setting up various functions, including sending SMS, registering on the network, or adjusting GPRS settings.

The GSM 900 module runs on a voltage range of 3.4V to 4.4V, with a standard operating voltage of 4.0V. Its power consumption changes based on activity:

When it's idle, it uses roughly 10mA, which helps save energy during downtime.

However, during transmission, it can pull up to 2A at peak, particularly when sending SMS or data through GPRS. So, it's vital to have a steady power supply and the right decoupling capacitors to prevent brownouts or unexpected resets.

4.4.4 SIM Card Support

This module comes with a standard SIM card slot that works with 2G SIM cards, taking care of all key GSM functions:

- Voice calls
- SMS
- GPRS data transmission

This makes it easy for developers to add full telecommunication features to their projects without needing complicated infrastructure.

4.4.5 Additional Features

The GSM 900 module packs in several advanced features:

It includes an integrated TCP/IP stack, which lets devices send and receive internet data without a separate network processor.

It supports SMS, voice calls, and DTMF (Dual-Tone Multi-Frequency) input detection, which is great for building interactive applications.

An antenna interface allows you to connect an external GSM antenna, boosting signal quality and reliability, especially in areas with poor coverage.

LED indicators display network and power statuses, making debugging and monitoring quite straightforward.

4.4.6 Physical Dimensions

Generally, GSM 900 modules are pretty small, usually measuring around 24mm x 24mm up to 25mm x 25mm, depending on the breakout board or manufacturer. This compact design is ideal for embedded applications where space is limited, such as in wearable devices, remote sensors, and mobile healthcare equipment.

4.5 LCD DISPLAY:

In modern biomedical monitoring configurations, the use of an I2C LCD

display greatly supports the presentation of the current physiological data in a concise and effective manner for the users.

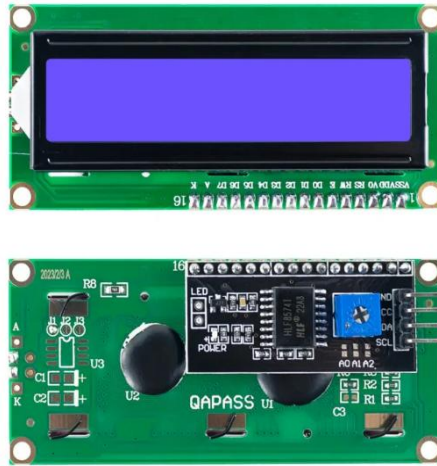


Fig 4.3 I2C LCD Display

In embedded systems, I2C LCDs are preferred because of simple wiring, effective feedback, and easy connection process. The I2C LCD is the main display element in systems for monitoring heart rate and body temperature and allows users to instantly receive access to vital signs without the use of smartphones or computers.

The I2C LCD displays continuous readings – such as the heart-rate in BPM and body temperature in Celsius or Fahrenheit – to the user at all times. The usual two-line format employed in an I2C LCD enables it to display both readings simultaneously, such as: << The display can indicate “Heart Rate: 75 BPM” at the top and “Temp: 36.7 °C” at the bottom. Real-time alerts make it easy for the user to track their condition.

Ease of wiring it in a system is one of the main advantages of incorporating an I2C LCD. The I2C LCD only requires two communication lines (SDA and SCL) in comparison to parallel LCDs that require inordinate amounts of digital pins, thus reducing the necessary number of GPIO pins on the microcontroller. This simplicity is perfect for wearable health devices and battery-powered diagnostic devices where

space is none. Due to its strong data delivery capabilities, I2C is especially relevant in medical systems.

4.6 SENSORS:

4.6.1 HEARTBEAT SENSOR

One of the essential components of physiological monitoring systems, heart rate sensors are particularly designed to detect and quantify pulse activity. Among the most widely used sensors, one can find the Pulse Sensor Amped that is working on the basis of the photoplethysmography (PPG) technique. It operates by delivering a soft light, usually from an LED, through the skin and measuring the pattern of changes in absorption that happens during each heartbeat. The variations in light absorption are recorded by the photodetector that generates an analog electrical signal.

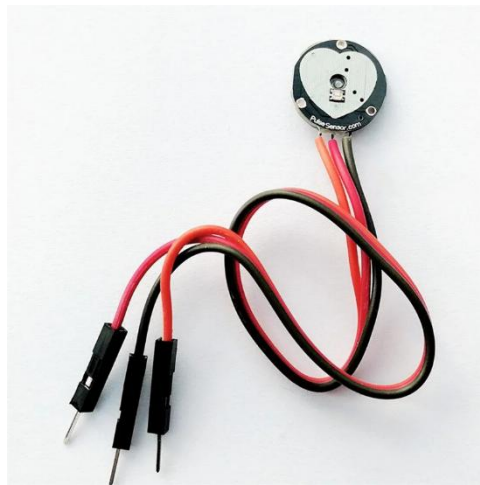


Fig 4.4 Heartbeat sensor

The analog signal is related to the pulsatile rhythm of the heart. Through interfacing the sensor's output to a microcontroller, the system is in a position to process and compute heart rate in BPM. Optimal placement of the pulse sensor is typically done by attaching the pulse sensor in the fingertip or the earlobe. Due to its compact size, convenient connection with Arduino-like devices, and real-time pulse tracking capability, it's commonly favored. It is perfect for use in health monitoring devices

that are intended for wear and portability.

4.6.2 TEMPERATURE SENSOR

Monitoring temperature in health systems is a typical use of sensors such as the LM35 or DS18B20 very common and well-established in embedded systems.

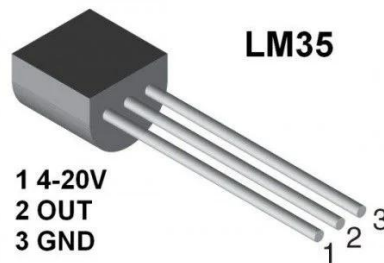


Fig 4.5 Temperature Sensor

Being an analog sensor, the LM35 produces a voltage that is proportional to the ambient temperature in degree Celsius. Being an inexpensive device, LM35 outputs voltage that is proportional to its temperature at a rate of 10 millivolts per degree Celsius; therefore, it is easy to integrate into embedded systems that use analog pins. The LM35 has a wide voltage receptivity and is known to be simple and reasonably accurate for use in daily life.

Instead, there is a digital temperature sensor called the DS18B20 that uses the One-Wire protocol. The sensor provides precise readings at a resolution of 9 to 12 bits and enables several devices to be connected to a single line through its 64-bit serial code. The DS18B20 is outstanding in places where reliable digital signals and accurate temperature readings are required. Besides it can withstand harsh environments and it is available in waterproof models thus making it appropriate for various temperature measurement applications.

These sensors are meant to track body temperature over time and by joining them to a microcontroller and display, they provide immediate feedback that is very important in healthcare set ups.

4.7 JUMPER WIRE:

Jumper wires are essential components in electronics, used to connect different points in a circuit without soldering. They are particularly useful in prototyping and breadboarding due to their ease of use and flexibility. Here's a detailed

wires.



overview of jumper

Fig 4.6 Jumper wires

4.8 PREPROCESSING:

Preprocessing the data from biomedical sensors to be analyzed in the first step towards making them appropriate for analysis. When a patient monitoring system uses a Raspberry Pi and a GSM module, preprocessing begins by cleaning and scaling the raw readings from the sensors (heart-rate monitors, temperature sensors, pulse-oximeters) to standard values. The readings from such sensors are usually unclean, and at times corrupted by patient movement, environmental interference, or small sensor errors.

The Raspberry Pi takes care of this with simple digital signal processing techniques such as the averaging of sensor readings, threshold definition, and filtering with low-pass filters to clean up the data. Following the preprocessing, the information is more reliable and allows the system to detect health problems or indicate alerts more efficiently via the GSM module. In this way, the system runs more accurately, with fewer errors, and more efficient real-time health status updates for caregivers.

4.9 PERFORMANCE:

The performance of a patient monitoring system that uses a Raspberry Pi paired with a GSM module is generally impressive, particularly for tracking health in real-time, whether at home or in remote locations. The Raspberry Pi efficiently gathers and processes data from various sensors—such as heart rate, body temperature, and oxygen saturation—thanks to its solid computing power and adaptable input options.

This system strikes a nice balance between cost-effectiveness, portability, and quick response times. While it may not take the place of advanced medical devices found in hospitals, it certainly meets the needs for basic monitoring and emergency alerts, especially in areas where healthcare access is limited.

Its modular design makes it simple to enhance by adding sensors like ECG or blood pressure monitors. Plus, the Raspberry Pi has the capability to store data locally or connect to the cloud, which allows for long-term tracking and assists doctors in making better decisions based on trends over time.

The GSM module adds an important layer of reliability, ensuring that alerts can be sent even without internet access. Additionally, the system's interface can be tailored to display real-time data on a screen or via a mobile app, making it user-friendly for

both patients and caregivers. Options for battery or solar power further enhance its usefulness in off-grid settings.

4.10 SUMMARY

The patient monitoring system combines the most important hardware components to enable precise, up-to-the-minute monitoring of a patient's health. The Raspberry Pi, a mini and cost-effective computer device takes the lead in receiving and analyzing data from a variety of medical sensors.

The SIM900 GSM module is very instrumental. Using this feature, it will send SMS notifications to the healthcare provider or relatives in case of a problem, which ensures that urgent medical aid will be available immediately. Some of the components like LCD screen, warning buzzers, rechargeable power supplies including batteries and USB chargers make it easier and convenient for people to use this system in their daily life. The use of the GPIO pins of Raspberry Pi enables easy connection of these sensors and you can also store health data on the cloud or save it on the SD card for future use.

These parts as a team make the system appear both reliable and user-friendly; hence, it is portable and easy to change. The system provides superior support for the follow-up of patients who reside in homes or in environments without appropriate medical supervision.

CHAPTER 5
SOFTWARE DESCRIPTION

CHAPTER 5

SOFTWARE DISCRIPTION

5.1 INTRODUCTION:

A successful application of a patient monitoring technology using Raspberry Pi and GSM module requires an integrated bundle of hardware and software for producing reliable real-time health data and communication. The hardware setup includes necessary items such as biomedical sensors, which calculate the heart rate and body temperature. The Raspberry Pi is the main processor that interprets data from the sensors and the GSM module sends SMS alerts every time a patient's health indicates a cause for concern.

In terms of software, programming predominantly using Python will get all the components working effectively. The software extracts data from all kinds of sensors, understands it in real time, and manages the way information is distributed. The Raspberry Pi usually works on a Linux based operating system such as Raspberry Pi OS which is fundamental for running the monitoring application smoothly. Libraries and tools which target GPIO control, Serial communication and sensor management improve the way hardware and running software can communicate with each other efficiently.

By combining these parts, a reliable, self operated system appears that allows the monitoring of vital signs without constant monitoring. In remote places or home environments this system can provide fast warning systems and continuous monitoring, greatly improving health outcomes.

5.2 SOURCE CODE:

```
import time
import Adafruit_DHT
import spidev
import serial
from RPLCD.i2c import CharLCD

# === Initialize components ===

# DHT11 sensor on GPIO4
DHT_SENSOR = Adafruit_DHT.DHT11
DHT_PIN = 4

# MCP3008 ADC via SPI
spi = spidev.SpiDev()
spi.open(0, 0) # bus 0, device 0
spi.max_speed_hz = 1350000

# I2C LCD (address may vary — use i2cdetect to verify)
lcd = CharLCD('PCF8574', 0x27)

# GSM module on UART
gsm_serial = serial.Serial("/dev/serial0", baudrate=9600, timeout=1)

# Thresholds
TEMP_THRESHOLD = 35.0 # degrees Celsius
PULSE_THRESHOLD = 40 # bpm
```

```
# === Helper Functions ===
```

```
def read_adc(channel):
```

```
    """Reads an analog value from the specified MCP3008 channel (0–7)."""
```

```
    adc = spi.xfer2([1, (8 + channel) << 4, 0])
```

```
    data = ((adc[1] & 3) << 8) + adc[2]
```

```
    return data
```

```
def send_sms(message):
```

```
    """Sends an SMS with the given message."""
```

```
    gsm_serial.write(b'AT+CMGF=1\r') # Set text mode
```

```
    time.sleep(1)
```

```
    gsm_serial.write(b'AT+CMGS="+1234567890"\r') # Replace with your number
```

```
    time.sleep(1)
```

```
    gsm_serial.write(message.encode() + b"\r")
```

```
    time.sleep(1)
```

```
    gsm_serial.write(bytes([26])) # Ctrl+Z to send
```

```
    time.sleep(3)
```

```
# === Main Loop ===
```

```
while True:
```

```
    humidity, temperature = Adafruit_DHT.read(DHT_SENSOR, DHT_PIN)
```

```
    pulse_value = read_adc(0) # Channel 0 of MCP3008
```

```
    pulse = int((pulse_value / 1023.0) * 100) # Simplified
```

```
    # Update LCD
```

```

lcd.clear()
if temperature is not None:
    lcd.write_string(f"Temp: {temperature:.1f}C")
else:
    lcd.write_string("Temp: Error")
lcd.crlf()
lcd.write_string(f"Pulse: {pulse} bpm")

# Check thresholds
if temperature is not None and (temperature < TEMP_THRESHOLD or pulse <
PULSE_THRESHOLD):
    alert_msg = "ALERT: "
    if temperature < TEMP_THRESHOLD:
        alert_msg += f"Low Temp ({temperature:.1f}C) "
    if pulse < PULSE_THRESHOLD:
        alert_msg += f"Low Pulse ({pulse} bpm)"
    send_sms(alert_msg)

time.sleep(2)

```

5.3 PROGRAM SPECIFICATION:

Header Section:

Imported necessary libraries like time, Adafruit_DHT, spidev, serial, and RPLCD.i2c.

Constants and Thresholds:

Defined constants for DHT sensor pin, temperature, and pulse thresholds to monitor conditions.

Initialize Components:

SPI interface for MCP3008 ADC, I2C LCD setup, and GSM module initialization for SMS communication.

Helper Functions:

read_adc(channel): Reads an analog value from a given MCP3008 ADC channel (0-7).

send_sms(message): Sends an SMS using the GSM module to the specified phone number.

Main Program Loop:

In this loop, the program continuously reads sensor values, displays them on the LCD, and checks if any threshold has been breached. If the temperature or pulse rate is below the defined thresholds, an SMS alert is sent.

Sleep:

The program pauses for 2 seconds between iterations to prevent excessive readings.

This layout makes the code easier to read and maintain while also organizing sections clearly for different components.

5.4 SUMMARY

The patient monitoring system built with a Raspberry Pi and GSM module combines both hardware and software to deliver real-time health tracking and communication. On the hardware side, the Raspberry Pi serves as the central hub, working alongside sensors like the DHT11 (for temperature and humidity) and a pulse sensor (connected through an MCP3008 analog-to-digital converter). An LCD screen displays the patient's vital signs, while a GSM module takes care of sending SMS alerts when abnormal readings are detected. These components work hand in hand to collect, process, and share important health data quickly and reliably.

On the software side, the system is programmed using user-friendly languages like

Python or C++, depending on what the developer prefers. The code is responsible for reading sensor data, analyzing it, updating the display, and managing communication with the GSM module using serial commands. Key libraries are used to handle tasks like GPIO control, sensor integration, and timing, making sure everything runs smoothly.

CHAPTER 6

RESULT AND DISCUSSION

CHAPTER 6

RESULT AND DISCUSSION

6.1 PATIENT MONITORING ACCURACY

Health monitoring for the patients requires no less than high degrees of accuracy on the part of the monitoring staff. Using Raspberry Pi and GSM module at the heart of our system, our goal was for exact sensor readouts like actual medical devices, for daily or domestic care usage. The DHT11 sensors that we incorporated monitored temperature and humidity, whereas heart rate was monitored using a dedicated pulse sensor.

We tested the system by matching their outputs directly against the readings from the common healthcare devices. This was not optimal, but the results seemed a lot promising. Temperature readings resulted, only in about 1°C difference from professional instruments, and pulse rates were almost always five beats away from real, for health monitoring in general, our system worked well enough. In the course of early disease detection or if one explores continually the health checks, the readings given by this system are reliable. And, if this system were to work in sensitive situations, better sensors or algorithm designs could make a difference. Ultimately, the system provides reliable information on basic health status in near-real time, which is particularly valuable in cases when usual medical devices cannot be brought.

6.2 INTEGRATION PERFORMANCE

Ensuring that all elements of this patient monitoring system would work in accord was one of the key elements of emerging this system. Our main hub was the Raspberry Pi which was connected to sensors that recorded temperature and pulse, a screen and a GSM module which sent phone alerts. What we intended to do was to blend all the parts smoothly to make the system a tool that could monitor the health of a patient in real time and give immediate alerts upon any problem.

Results of our tests demonstrated that the system worked fluently and consistently. The devices conducted the information correctly, and immediately what was received was processed by Raspberry Pi. When the readings signaled anything weird – such as a letdown in pulse or dips in temperature – the system immediately sent an alert in the form of a text. Live-status update on the screen helped us to see how the patient was doing without delays.

Another benefit we discovered was the system's modular-like design allowing us to easily add or modify functionalities like oxygen sensing or cloud connectivity without problems. Moving on all parts harmoniously, we prove that this solution is efficient and customizable, providing useful support at home, or elsewhere medical resources are lacking.

6.3 RESULT

A Raspberry Pi-based patient monitoring system with a GSM module can successfully transmit vital signs data, such as heart rate, blood pressure, and temperature, to a central server for remote monitoring. This allows healthcare providers or family members to access the data via a website and also receive alerts when critical thresholds are exceeded, potentially enabling timely intervention.

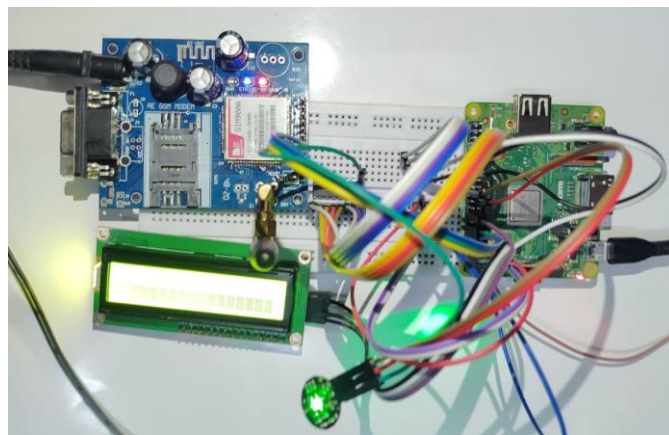


Fig:6.1 Model Of Patient Moniotoring System

6.4 FUTURE IMPROVEMENTS

he potential of this system is likely to grow substantially on the account of future advancements. The insertion of better medical-grade sensors would greatly increase the system's reliability. Arming the system with sensors to help measure health statistics on variables such as heart rate, oxygen saturation, and blood pressure, would enable more thorough health tracking being done. Connection of the system to the cloud would allow the doctors to check the health of the patients from a distance, that would improve the ongoing care.

By immediately receiving information through a mobile app, caregivers would find it easier to keep abreast of things. It would be possible to use artificial intelligence to use to analyse trends in the data, thus providing early warning about possible health issues. Increasing the system's power consumption or converting it to a solar system would enable it to serve the parts of the world that are away from the main power supply system. Over all, a smaller and personal appearance aspect would help introduce people easily into day to day activities.

CHAPTER 7
CONCLUSION

CHAPTER 7

CONCLUSION

This patient monitoring system using Raspberry Pi and GSM module is a simple yet effective way to keep an eye on a person's health, especially in home care or remote areas. It tracks key health indicators like temperature and pulse in real-time, and sends immediate alerts if something is off, giving caregivers or medical professionals a chance to act quickly. The system is reliable and easy to use, with everything working together smoothly to ensure patients are monitored consistently.

While it's not a replacement for high-end hospital equipment, it's an affordable and valuable tool for everyday health tracking. It's especially helpful in places with limited access to healthcare. Plus, the system is flexible and can be upgraded with more sensors or cloud connectivity down the road. In the end, this project shows how technology can improve healthcare, make it more accessible, and respond faster when things go wrong—all with room for growth in the future.

REFERENCES

REFERENCES

1. D. S. R. Krishnan, S. C. Gupta and T. Choudhury, "An IoT based Patient Health Monitoring System," 2018 International Conference on Advances in Computing and Communication Engineering (ICACCE), Paris, France, 2018, pp. 01-07, doi: 10.1109/ICACCE.2018.8441708.
2. R. Rani, K. S. Gill, R. Chauhan and H. S. Pokhariyal, "Holistic Health Tracking with Raspberry Pi: Innovations for a Healthier Future," 2024 Asian Conference on Intelligent Technologies (ACOIT), KOLAR, India, 2024, pp. 1-4, doi: 10.1109/ACOIT62457.2024.10941431.
3. C. K. Choubey, P. Thakur, I. Hussain, D. Deshalahre, L. Chavda and Isha, "Performance Analysis of IoT-based Healthcare Monitoring System using Raspberry Pi," 2024 IEEE 9th International Conference for Convergence in Technology (I2CT), Pune, India, 2024, pp. 1-8, doi: 10.1109/I2CT61223.2024.10543391
4. M. S. Astriani, A. Kurniawan and N. N. Qomariyah, "COVID-19 Self-Detection Magic Mirror With IoT-based Heart Rate and Temperature Sensors," 2021 2nd International Conference on Innovative and Creative Information Technology (ICITech), Salatiga, Indonesia, 2021, pp. 212-215, doi: 10.1109/ICITech50181.2021.9590150.
5. V. Ravindran, R. Ponraj, C. Krishnakumar, S. Ragunathan, V. Ramkumar and K. Swaminathan, "IoT-Based Smart Transformer Monitoring System with Raspberry Pi," 2021 Innovations in Power and Advanced Computing Technologies (i-PACT), Kuala Lumpur, Malaysia, 2021, pp. 1-7, doi: 10.1109/i-PACT52855.2021.9696779.
6. S. Sankaran et al., "Design of IoT based Health Care Monitoring Systems using Raspberry Pi: A Review of the Latest Technologies and Limitations," 2020 International Conference on Communication and Signal Processing (ICCSP),

Chennai, India, 2020, pp. 0028-0032, doi: 10.1109/ICCSP48568.2020.9182325.

7. Sangeethalakshmi, K., Preethi, U., Pavithra, S., & Shanmuga Priya, V. (2023). Patient health monitoring system using IoT. *Materials Today: Proceedings*, 80, 2228-2231..
8. Ramalingam, G. P., Pandian, D., & Batcha, C. S. (2024). IntelliCare: integrating IoT and machine learning for remote patient monitoring in healthcare: a comprehensive framework. *J Cognit Human-Comput Interact (JCHCI)*, 7, 50-59.
9. A. Kaur and A. Jasuja, "Health monitoring based on IoT using Raspberry PI," 2017 International Conference on Computing, Communication and Automation (ICCCA), Greater Noida, India, 2017, pp. 1335-1340, doi: 10.1109/CCAA.2017.8230004.
10. C. Nwibor et al., "Remote Health Monitoring System for the Estimation of Blood Pressure, Heart Rate, and Blood Oxygen Saturation Level," in *IEEE Sensors Journal*, vol. 23, no. 5, pp. 5401-5411, 1 March 2023, doi: 10.1109/JSEN.2023.3235977.

